David Cyncynates

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Employment

2025–Present **Postdoctoral Fellow** International Center for Theoretical Physics, Trieste, IT
Supervisor: Prof. Giovanni Villadoro

2022–2025 **Postdoctoral Scholar** University of Washington, WA, USA
Supervisor: Prof. Masha Baryakhtar

Education

2017–2022 **Ph.D. Theoretical Physics** Stanford University, CA, USA
Advisor: Prof. Savas Dimopoulos.
Thesis: "*Strongly Nonlinear Axion Dynamics*"

2013–2017 **B.Sc. Mathematics & Physics** *summa cum laude* Case Western Reserve University, OH, USA
Thesis: "*Reconsidering Seismological Constraints on the Available Parameter Space of Macroscopic Dark Matter*," under supervision of Prof. Glenn D. Starkman.

Research Interests

Description I am a theoretical physicist specializing in Beyond the Standard Model (BSM) particle phenomenology. My recent work focuses on identifying well-motivated experimental targets for vector and scalar dark matter, where theoretical priors are relatively weak compared to more established candidates like the QCD axion. My earlier research includes publications on the QCD axion, axion-like particles, nontopological solitons, and macroscopic dark matter.

Keywords **hep-ph, astro-ph.co**

Publications & Products

Preprints

Pouya Asadi, David Cyncynates, Stefania Gori, *Technically natural baryogenesis with the QCD axion*, `arXiv:2610.20706`.

Masha Baryakhtar, David Cyncynates, Ella Henry, *Axiverse Lampposts*, `arXiv:2602.23424`.

Pouya Asadi, David Cyncynates, Stefania Gori, *Axiverse Baryogenesis*, `arXiv:2511.15794`.

Refereed

David Cyncynates, Olivier Simon, *Scalar relics from the hot Big Bang*, Phys. Rev. Lett **135** (2025).

David Cyncynates, Olivier Simon, *Minimal targets for dilaton direct detection*, Phys. Rev. D **112** (2025).

Editors' Suggestion David Cyncynates, Zachary Weiner, *Experimental targets for dark photon dark matter*, Phys. Rev. D **111** (2025).

David Cyncynates, Zachary Weiner, *Detectable, defect-free dark photon dark matter*, Phys. Rev. Lett **134** (2025).

David Cyncynates, Jedidiah Thompson, *Heavy QCD axion dark matter from avoided level crossing*, Phys. Rev. D **108** (2023).

David Cyncynates, Olivier Simon, Jedidiah Thompson, Zachary Weiner, *Nonperturbative structure in coupled axion sectors and implications for direct detection*, Phys. Rev. D **106** (2022).

David Cyncynates, Tudor Giurgica-Tiron, Olivier Simon, Jedidiah Thompson, *Resonant nonlinear pairs in the axiverse and their late-time direct and astrophysical signatures*, Phys. Rev. D **105** (2022).

David Cyncynates, Tudor Giurgica-Tiron, *Structure of the oscillon: The dynamics of attractive self-interaction*, Phys. Rev. D **103** (2021).

David Cyncynates, Joshua Chiel, Jagjit Sidhu, and Glenn D. Starkman, *Reconsidering seismological constraints on the available parameter space of macroscopic dark matter*, Phys. Rev. D **95** (2017).

Software

David Cyncynates, *Axiverse machine GitHub repository,* https://github.com/DavidCyncynates/Axiverse_Machine, (2024)

David Cyncynates, *Superradiance GitHub repository,* https://github.com/DavidCyncynates/Superradiance, (2024)

David Cyncynates, Tudor Giurgica-Tiron, *Simple oscillon GitHub repository,* https://github.com/SimpleOscillon/Code, (2021).

Seminars & Conference Presentations

21 Oct. 2026 **Invited Seminar** *Super Rad Reverberations.* IFAE, Barcelona, ES.

1 July. 2026 **Invited Talk** *Axiverse Lampposts.* Sanford Underground Research Facility, South Dakota, USA.

24 June. 2026 **Invited Talk** *Axiverse Lampposts.* Cosmic WISPers, Corfu, GR.

15 June. 2026 **Invited Talk** *Axiverse Lampposts.* MITP, Mainz, DE.

23 Apr. 2026 **Invited Seminar** *Axiverse Lampposts.* CERN Theory BSM Forum, Geneva, CH.

19 Mar. 2026 **Invited Seminar** *Axiverse Lampposts.* IFAE, Barcelona, ES.

11 Mar. 2026 **Invited Seminar** *Axiverse Lampposts.* University of Padova, Padova, IT.

3 Feb. 2026 **Invited Seminar** *Axiverse Baryogenesis.* University of Washington, Seattle, WA, USA.

29 Jan. 2026 **Invited Seminar** *Axiverse Baryogenesis.* Texas A&M, College Station, TX, USA.

28 Jan. 2026 **Invited Seminar** *Detectable dark photon dark matter.* Rice, Houston, TX, USA.

23 Jan. 2026 **Invited Seminar** *Axiverse Baryogenesis.* LITP, Stanford, CA, USA.

21 Jan. 2026 **Invited Seminar** *Axiverse Baryogenesis.* UCSC, Santa Cruz, CA, USA.

12 Jan. 2026 **Invited Seminar** *Detectable dark photon dark matter.* Indiana University, Bloomington, IN, USA.

16 Nov. 2025 **Contributed Talk** *Baryogenesis in the Axiverse.* Axions in Japan 2025, IPMU, Tokyo University, Tokyo, Japan.

9 Nov. 2025 **Contributed Talk** *Baryogenesis in the Axiverse.* TeVPA 2025, University of Valencia, Valencia, ES.

9 June. 2025 **Invited Talk** *Scalar relics from the hot big bang.* Sanford Underground Research Facility, Lead, SD, USA.

14 May. 2025 **Contributed Talk** *Scalar relics from the hot big bang.* Tabletop Scale Cosmology, Online.

9 May. 2025 **Contributed Talk** *Learning our place in the Axiverse.* Berkeley Axion Workshop 2025, LBNL, Berkeley, CA, USA.

25 Feb. 2025 **Invited Seminar** *Scalar relics from the hot big bang.* Case Western Reserve University, Cleveland, OH, USA.

4 Feb. 2025 **Invited Seminar** *Scalar relics from the hot big bang.* UCSD, San Diego, CA, USA.

14 Jan. 2025 **Workshop Discussion** *(Un)factorizable axiverse.* Prospects for the String Axiverse, BIRS, Banff, CA.

29 Nov. 2024 **Invited Seminar** *Experimental targets for vector dark matter.* Cosmic WISPers November Meeting, Europe.

1 Nov. 2024 **Workshop Talk** *Scalars remember the hot big bang.* Kashiwa-no-ha Dark Matter and Cosmology Symposium, IPMU, Tokyo University, Tokyo, Japan.

22 Oct. 2024 **Contributed Poster** *Experimental targets for vector dark matter.* Cosmo '24, Kyoto, Japan.

26 Aug. 2024 **Contributed Talk** *Scalars remember the hot big bang.* TeVPA 2024, University of Chicago, Chicago, IL, USA.

10 Jul. 2024 **Invited Seminar** *Scalars remember the hot big bang.* Perimeter Institute for Theoretical Physics, Waterloo, ON, CA.

29 Apr. 2024 **Invited Seminar** *Detectable Vector Dark Matter.* CENPA, Seattle, WA, USA.

22 Apr. 2024 **Invited Seminar** *Detectable Vector Dark Matter.* IFS, Eugene, OR, USA.

17 Apr. 2024 **Invited Seminar** *Detectable Vector Dark Matter.* TRIUMF, Vancouver, BC, CA.

2 May. 2023 **Invited Talk** *Detectable Dark Photon Dark Matter.* LCTP, Ann Arbor, MI, USA.

1 Mar. 2023 **Invited Seminar** *Detectable Dark Photon Dark Matter.* LBNL, Berkeley, CA, USA.

21 Feb. 2023 **Invited Seminar** *Detectable Dark Photon Dark Matter.* University of Washington, Seattle, WA, USA.

27 Nov. 2022 **Invited Workshop Talk** *Cosmology with Coupled Axions.* IBS-ICTP-MultiDark Workshop, IBS Daejeon, Korea.

14 Jul. 2022 **Contributed Talk** *Long-Lived Exotic Objects from Nearly-Free Theories.* University of Washington, Seattle, WA, USA.

18 Mar. 2022 **Invited Seminar** *Structure of the Oscillon.* University of Nottingham, Nottingham, GB-ENG.

2 Mar. 2022 **Invited Seminar** *Friendship in the axiverse.* Arizona State University, Tempe, AZ, USA.

15 Feb. 2022 **Invited Seminar** *Friendship in the axiverse.* String Phenomenology Seminar Series.

18 Jan. 2022 **Invited Seminar** *Friendship in the axiverse.* Case Western Reserve University, Cleveland, OH, USA.

8 Jan. 2022 **Contributed Poster** *String theory motivates strongly nonlinear classical dynamics in the early universe.* Dynamics Days, Georgia Tech, Atlanta, GA, USA.

30 Nov. 2021 **Invited Seminar** *Friendship in the axiverse.* Perimeter Institute for Theoretical Physics, Waterloo, Ontario, CA.

23 Nov. 2021 **Local Seminar** *Friendship in the axiverse.* University of Washington, Seattle, WA, USA.

9 Nov. 2021 **Invited Seminar** *Friendship in a dense axiverse.* Cornell, Ithaca, NY, USA.

1 Nov. 2021 **Invited Seminar** *Friendship in a dense axiverse.* Georg-August-Universität, Göttingen, Niedersachsen, DE.

18 Oct. 2021 **Invited Seminar** *Friendship in the axiverse.* BCTP, Berkeley, CA, USA.

1 Oct. 2021 **Invited Seminar** *The friendly axiverse.* NYU, New York, NY, USA.

18 Jun. 2021 **Local Seminar** *Axiverse dark matter.* SITP, Stanford, CA, USA.

20 May 2021 **Talk** *Will axions be discovered as oscillons? (Timestamp: 29:08)* XIV International Conference on Interconnections between Particle Physics and Cosmology at the University of Oklahoma, Norman, OK, USA.

4 May 2021 **Invited Seminar** *The Structure of the Oscillon: Form-Factor is Function.* Rice cosmology group, Houston, TX, USA.

9 Mar. 2018 **Local Seminar** *Periodic Magnetized Slabs as Axion Dark Matter Detector.* SITP, Stanford, CA, USA.

17 May 2017 **Conference Talk** *Reconsidering seismological constraints on the available parameter space of macroscopic dark matter.* The International Workshop on Baryon and Lepton Number Violation, Cleveland, OH, USA.

Spring 2017 **Local Poster** *Reconsidering seismological constraints on the available parameter space of macroscopic dark matter.* Intersections Symposium, Cleveland, OH, USA.

Spring 2016 **Local Poster** *Reconsidering seismological constraints on the available parameter space of macroscopic dark matter.* SOURCE Symposium, Cleveland, OH, USA.

Summer 2014 **Local Poster** *Constructing a toy model to explore the possible instability in Friedmann Robertson Walker spacetime.* SOURCE Symposium, Cleveland, OH, USA.

Computer Skills

- **Scientific Computing**
  Use of the Sherlock cluster at Stanford, the Symmetry cluster at the Perimeter Institute for Theoretical Physics, and the High Performance Computing Cluster at Case Western Reserve.
- **3+1D Cosmological Simulations**
  - Programmed pseudospectral RK4 integrator for multi-field cosmological simulations.
  - Analyzed simulation output for the formation of self-bound structures.
- **3D Ray Tracing**
  - Programmed seismological wave integrator for dark matter direct detection in the framework of the Preliminary Reference Earth and Moon Models.
  - Analyzed simulation output and estimate seismometer response.
- **Languages**
  Julia, Mathematica, Matlab, Python.

Teaching Experience

Winter 2022 **TA** *Mechanics*

Winter 2021 **TA** *Electricity, Magnetism, and Optics Laboratory*

Autumn 2020 **TA** *Light and Heat Laboratory*

Winter 2020 **TA** *Electricity, Magnetism, and Optics Laboratory*

Autumn 2018 **Head TA** *Mechanics and Heat*

Summer 2018 **TA** *Mechanics and Heat*

Spring 2018 **TA** *Modern Physics*

Autumn 2016 **Private Tutor** *Thermodynamics and Statistical Mechanics*

Outreach

2022 Art of Science Exhibition

2021–2022 Stanford Future Advancers of Science and Technology (FAST) Mentor

2021 Art of Science Exhibition

2018–2019 Stanford Future Advancers of Science and Technology (FAST) Mentor

2019 SLAC Kids Night presenter

2018 SLAC Kids Night presenter

In the News

2025 *Phys.org* Escaping cosmic strings: How dark photons could finally work as dark matter